

ARCHITECTURE.md

SynChain Architecture

High-Level Flow

User ↓ Frontend (Next.js) ↓ FastAPI Backend ↓ Agent Orchestrator ↓
_____ Demand Agent Inventory Agent Logistics Agent Risk Agent
_____ ↓ Decision Engine ↓ Database ↓ Results Dashboard

Agent Responsibilities

Demand Agent

Inputs:

- Demand
- Product Data

Outputs:

- Demand Forecast

Inventory Agent

Inputs:

- Current Stock
- Demand Forecast

Outputs:

- Inventory Recommendation

Logistics Agent

Inputs:

- Warehouse Information
- Inventory Data

Outputs:

- Warehouse Recommendation

Risk Agent

Inputs:

- Supplier Delay
- Supply Constraints

Outputs:

- Risk Assessment

Decision Engine

Combines outputs from all agents and generates the final strategy.

Current Architecture Status

Implemented:

- Frontend
- Backend API
- Database
- Agent Structure

Planned:

- Agent Orchestrator
- Digital Twin State Engine

Future:

- External Intelligence Layer
- Real-Time Event Processing